Wilbert Andrew Yonathan

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SUMMARY

Artificial Intelligence Engineering student at Xiamen University Malaysia (CGPA: 3.77/4.0) specializing in computer vision and deep learning. Experienced in designing lightweight object detection models, optimizing CNN architectures, and deploying end-to-end AI applications using PyTorch, OpenCV, and Streamlit. Seeking an AI Engineering or Computer Vision internship to contribute to real-world machine learning solutions.

EDUCATION

Xiamen University Malaysia Sep. 2023 – Sep. 2027
B.Eng. in Artificial Intelligence (CGPA: 3.77 */ 4.0)* *Selangor, Malaysia*
Honors & Awards: Dean's List, XMUM Merit Scholarship (Awarded 3 consecutive years, 2023–2025)

RELEVANT COURSEWORK

Computer Vision, Deep Learning, Advanced Machine Learning, Natural Language Processing, Deep Reinforcement Learning, Big Data Analytics, Statistical Learning, Robot Kinematics and Dynamics.

SKILLS

Programming: Python, C, C++

AI & ML Technologies: PyTorch, TensorFlow, OpenCV, YOLO, CNNs

Data & Development Tools: NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, Streamlit

Languages: English, Mandarin, Malay, Indonesian

PROJECTS

DenseNet-121 Backbone Integration in YOLOv8n for Paddy Disease Detection 2026 – Present
- Designed and implemented a compressed DenseNet-121 backbone integrated into YOLOv8n, reducing model complexity and parameter count while maintaining competitive object detection performance.
- Curated and cleaned an 8,000+ image agricultural dataset, enforcing a leakage-free validation protocol to ensure reliable and unbiased model evaluation.

GhostConv and SimSPPF Integration for Marine Organism Detection 2026
- Optimized the DU-MobileYOLO architecture for marine organism detection on the URPC2020 dataset by improving the performance-to-complexity trade-off.
- Replaced the SPPCSPC module with SimSPPF and standard convolutions with GhostConv in the Multi-Concat block, reducing computational complexity while maintaining competitive detection performance.

ResNet Unified Convolution Network (Res_UCN) 2025–2026
- Optimized ResNet-18 with unified convolutions, achieving a 69% reduction in model parameters.
- Designed a custom Quinsh activation function to improve gradient stability, achieving 91.45% classification accuracy on CIFAR-10.

Face Expression Recognition (FER2013) 2025
- Developed a custom CNN for facial expression recognition, achieving 66% accuracy and outperforming ResNet50, MobileNetV2, and VGG16 under identical evaluation settings.
- Integrated OpenCV Haar Cascade into a real-time webcam inference pipeline for face detection and emotion recognition.

Alzheimer's Disease Prediction System 2024
- Developed a Streamlit-based web application for MRI image classification and interactive prediction visualization.
- Integrated pretrained deep learning models into an end-to-end inference pipeline for medical image analysis.

ACTIVITIES

Love in Action (Community Service) 2026
Head of Photography and Videography: Managed media workflows and edited outreach materials.

XMUM AI Club 2025
Head of General Affairs: Managed logistics and coordinated external collaborators for a national-level AI event.

AIESEC XMUM 2024
General Affairs Member: Optimized administrative workflows and improved internal communication efficiency.